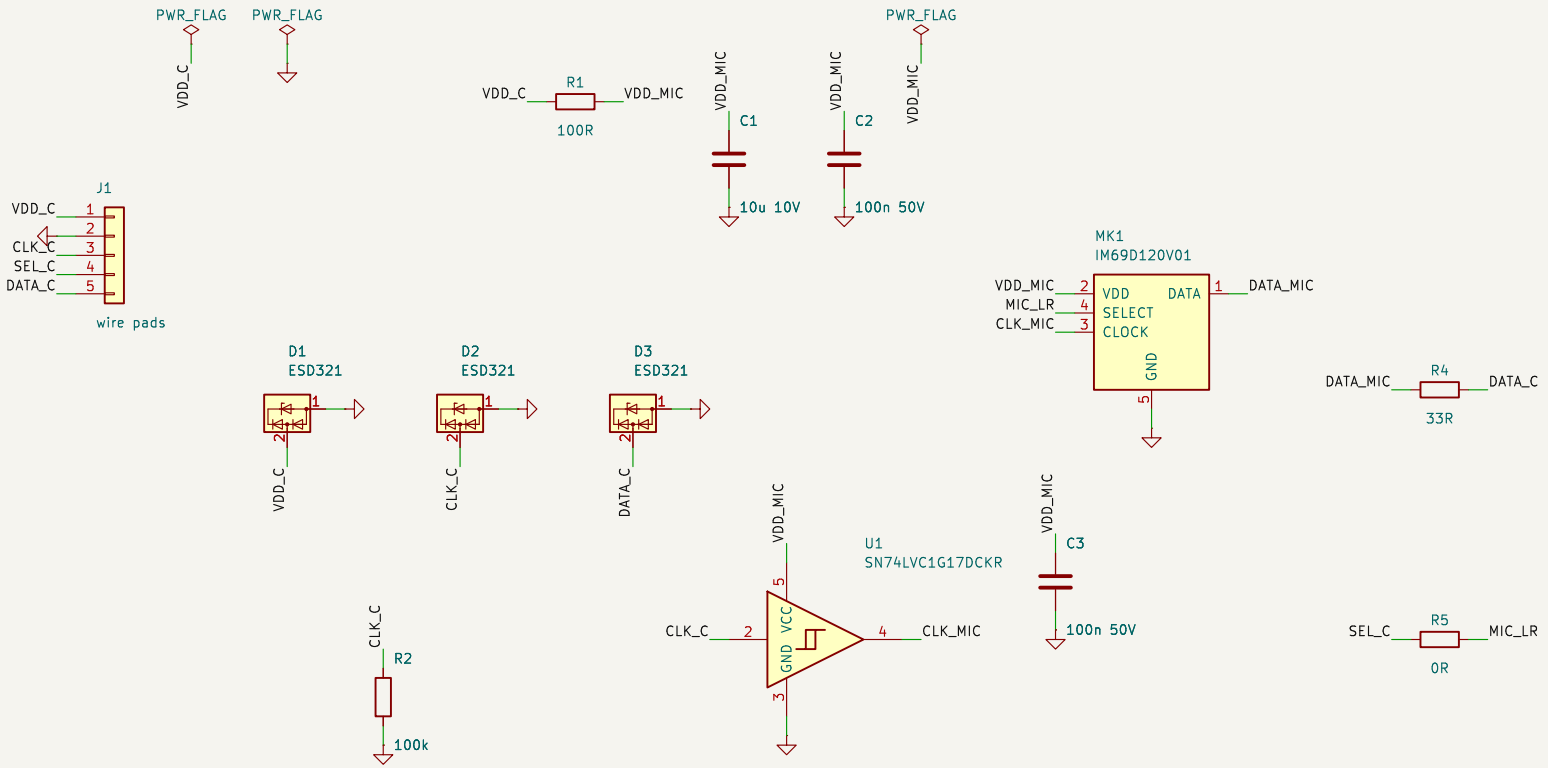
MIC STRIP (IM69D120 bottom-port PDM mic, one design for both strips; cable to the ReSpeaker Lite vacated U4/U5 pads)



IM69D120 (Infineon datasheet rev 1.0, 2017-12-20): VDD 1.62-3.6 V; 3.072 MHz lies in the 2.9-3.3 MHz operating range
  (no mode-entry restriction; start-up 20 ms), clock duty 48-52 % at >= 2.9 MHz, rise/fall 13 ns; IDD 980 uA typ;
  -26 dBFS, SNR 69 dB(A); ambient <= 70 C above VDD 3.0 V. SELECT has no stated internal pull: driven by the SEL wire via R5.
Cable (JST SR 1.0, A05SR05SR30K305B, both ends cut, soldered to J1): 1 VDD, 2 GND, 3 CLK, 4 SEL, 5 DATA - SEL is static,
  so it separates CLK from DATA like a ground. Default: U1 fitted, R4 33 Ohm, R5 0 Ohm.
Levers (bench 2.5): U1 bypass = U1 removed, its pads 2-4 bridged with wire (R3 dropped for space); R4 100 Ohm.
Test points: the wire pads (no TP1-TP4, decision 9.1).
Passives 0603 (D42); ESD in SOD-523 (D42: leaded alternative to the X1SON); MK1 LGA (no leaded MEMS exists).
Decision 8.1 (2026-11-07): SPH0641LU4H-1 dropped - it must not start in ultrasonic mode (3.072-4.8 MHz), where the XU316 runs.

Two boards of this design: L and R set by the cable SEL wire
Brief claude/brief-electronics-design-2026-11-07.md, K21, E0-D6
KiCad 9.0.9 - generated by hardware/main-board/gen/mic_strip.py (checkpoint 1)
HA panel (neocleous/ha-panel)

Sheet: /
File: mic-strip.kicad_sch

Title: HA panel mic strip

Size: A4	Date: 2026-11-07	**Rev: 1**
KiCad E.D.A. 9.0.9		Id: 1/1